THE RIGHT ADJOINT OF THE FIBRE PRODUCT FUNCTOR ON MODULE CATEGORIES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The general adjunction statement for the module category of a fibre product of rings. Lemma \ref{lemma-modules}, cited at proof lines 1017-1018. Gives the adjointness.
- The corresponding identification statements for the fibre product of modules. Lemma \ref{lemma-module-over-fibre-product}, cited at proof lines 1020-1023. Supply the two base change identifications after rewriting them.

Source reference. Original lemma 08KM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10890–10890. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08KK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10888–10888. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 08KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10889–10889. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D2F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16380–16380. Navigation pointer to a statement the

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Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In Situation 08KK the functor (08KL) has a right adjoint, namely the functor*

$$F : (N, M', \varphi) \longmapsto N \times_{\varphi, M} M'$$

where $M = M'/IM'$. Moreover, the composition of F with (08KL) is the identity functor on $Mod_D \times_{Mod_C} Mod_{C'}$. In other words, setting $N' = N \times_{\varphi,M} M'$ we have $N' \otimes_{D'} D = N$ and $N' \otimes_{D'} C' = M'$.

Proof. The adjointness statement follows from the more general Lemma 0D2F. The final assertion follows from the corresponding assertion of Lemma 07RU because $N' \otimes_{D'} D = N' \otimes_{D'} D' \otimes_{B'} B = N' \otimes_{B'} B$ and $N' \otimes_{D'} C' = N' \otimes_{D'} D' \otimes_{B'} A' = N' \otimes_{B'} A'$. □